RISHIKESH OMBASE

Agentic AI Architect | Senior Data Scientist

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PROFESSIONAL SUMMARY

Agentic AI Architect and Senior Data Scientist with 6+ years of experience designing and delivering machine learning and generative AI systems, with deep domain expertise in financial services and trading data. Specialises in multi-agent orchestration, Model Context Protocol (MCP) tool integration, retrieval-augmented generation (RAG) and production LLMOps. Brings an evaluation-driven approach to every design, with guardrails, observability and human-in-the-loop controls for high-stakes workflows, and a track record of turning document-, voice- and data-heavy processes into reliable AI services.

CORE COMPETENCIES

Agentic AI Architecture • Multi-Agent Orchestration • Model Context Protocol (MCP) • Retrieval-Augmented Generation (RAG) & GraphRAG • LLM Evaluation & Guardrails • LLMOps & Observability • Voice AI & Speech • Document AI & Vision-Language Models • Recommendation Systems • Financial Data & Quantitative Modelling • Cloud AI Platforms

PROFESSIONAL EXPERIENCE

Senior Data Scientist, **AIT Global India** **Dec 2023 – Present**

- Architected and delivered LLM-integrated API services and automated pipelines that extract structured insights from financial reports, combining OCR, Python and SQL.
- Built generative AI modules for summarising financial commentary and scoring sentiment across reports.
- Partnered on risk-scoring models enhanced with AI-augmented forecasting capabilities.
- Designed and deployed the API layer powering portfolio intelligence dashboards.

AI Data Specialist, **Futurense Technologies** **Jul 2023 – Oct 2023**

- Engineered real-time web data acquisition to enrich datasets for pricing models and market-trend analysis.
- Delivered executive decision dashboards using Python, Power BI and SQL.

Software Engineer, **Systech Solutions** **Jul 2022 – Mar 2023**

- Designed data frameworks for large-scale trading datasets on AWS.
- Enabled real-time updates to the analytics dashboards used to monitor asset performance.

Data Scientist, **Varilligence Technologies** **Jan 2020 – Mar 2022**

- Researched and developed machine learning models predicting stock-price movement using PySpark and financial indicators.
- Built alpha-generating trading signals from historical equities data, macroeconomic indicators and sentiment variables.
- Contributed to model validation and A/B testing of quantitative strategies in simulation environments.

SELECTED PROJECTS

FundDocs Studio – Agentic Document Styling for Fund Filings
Vision-language models, LLM agents, LangGraph, style-guide retrieval, structured outputs

- Designed a multi-agent pipeline that processes prospectuses, summary prospectuses and Statements of Additional Information (SAI) against each client's style guide.
- Vision-language models interpret page layout, a style-guide agent selects the governing rule for every heading, table and footnote, and LLM agents apply and verify formatting before routing documents for human sign-off.

ListingStudio – Multimodal Real-Estate Listing Generation

Vision LLMs, GPT models, FastAPI, prompt engineering, image classification

- Classified property images by room and feature, then generated listing content at scale with LLMs, including views and surroundings, amenities, neighbourhood details and full descriptions.
- Added a review step that validates generated claims against photos and listing data before publication.

CineMind – OTT Recommendation Engine

FastAPI, PostgreSQL, Redis, user embeddings, clustering, Docker, microservices

- Built the recommendation backend for an OTT platform that groups viewers into taste clusters from watch history, so users with similar habits receive similar titles; candidates are ranked per user.
- Used Redis to serve precomputed recommendations, sessions and trending lists, behind FastAPI microservices on PostgreSQL with authentication and access control.

EchoAgent – Real-Time Voice Agent

Whisper, streaming speech-to-text and text-to-speech, LLM function calling, WebSockets, FastAPI

- Built a conversational voice agent in which streaming speech-to-text feeds an LLM agent that calls tools to look up, book and update records, with streaming text-to-speech responses for low-latency replies.
- Handled mid-sentence interruptions and escalation to a human agent on request.

DubFlow – AI Video Dubbing Pipeline

Whisper, speaker diarization, LLM translation, text-to-speech, FFmpeg

- Built an end-to-end pipeline that dubs video into other languages: Whisper transcription with timestamps, speaker diarization, and context-aware LLM translation that preserves timing.
- Generated a text-to-speech voice for each speaker, aligned it to the original timing and mixed it back with the background audio.

AlphaDesk – Multi-Agent Quantitative Research Platform **In development**

LangGraph, MCP, GraphRAG, Qdrant, Neo4j, sandboxed code execution, Langfuse

- Hierarchical supervisor agent coordinating market-data, research, quantitative-analysis and risk agents, with human approval required before any output is released.
- Model routing across frontier and open-weight models, with full tracing and evaluation gates on every release.

Additional projects: invoice data automation (OCR, NLP); drone-type recognition (computer vision on Spark); FOX Channel BI data integration.

TECHNICAL SKILLS

Agent frameworks: LangGraph, CrewAI, OpenAI Agents SDK, Claude Agent SDK, Model Context Protocol (MCP), A2A

LLMs and models: Claude, GPT, Gemini, Llama, Qwen, DeepSeek, vision-language models; fine-tuning with LoRA, QLoRA and DPO

Speech and audio: Whisper, streaming speech-to-text and text-to-speech, speaker diarization, voice agents, video dubbing

Retrieval and memory: LlamaIndex, GraphRAG, hybrid search, reranking, Qdrant, pgvector, Neo4j, agent memory

LLMOps and governance: Langfuse, LangSmith, Ragas, DeepEval, guardrails, prompt-injection defence, OpenTelemetry

Engineering and cloud: Python, SQL, FastAPI, Pydantic, Docker, Kubernetes, Redis, PostgreSQL, vLLM, AWS (Bedrock, S3), Azure AI Foundry, Google Cloud Vertex AI

Machine learning: PyTorch, scikit-learn, time-series forecasting, classification, OpenCV, A/B testing

EDUCATION, CERTIFICATIONS & AWARDS

Master of Science, Computer Science, Shivaji University, Kolhapur **2022**

Bachelor of Science, Computer Science, Mahavir College, Kolhapur **2019**

Certifications: Machine Learning in Python; Data Analysis with Python; Data Visualization

Awards: Best Performer of the Month; Spot Award; GOM Award